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AZURE · DATA ENGINEERING · MEDALLION ARCHITECTURE

Cloud Data Pipeline Project on Microsoft Azure

Transforming raw, inconsistent data into actionable business insights through a scalable, end-to-end Medallion Architecture pipeline.

1,225+ Records Processed	**~95%** Data Quality Gain	**7** Error Types Fixed	**Parquet** Storage Format

01 - PROBLEM STATEMENT

Why This Pipeline Exists

In the digital era, organizations generate vast amounts of data from online platforms, applications, and user interactions. However, raw data is often incomplete, inconsistent, and unstructured, making meaningful analysis difficult without proper preprocessing.

Real-world datasets commonly contain multiple quality issues such as missing values, duplicate records, inconsistent labels, and outliers caused by manual input errors or fragmented data sources. Traditional processing methods often lack the scalability and automation required to clean and manage such data efficiently.

Another major challenge is the absence of an integrated system that unifies ingestion, transformation, storage, and visualization into a single automated workflow. Many organizations still rely on disconnected tools and manual processes, leading to inefficiency, human error, and unreliable analytics.

Critical Gaps Addressed:

- No unified platform for end-to-end data lifecycle management
- Limited automation in data cleaning and preprocessing
- Difficulty scaling analytics workflows with growing data volume
- Lack of separation between raw, clean, and aggregated data layers

02 - TECHNOLOGIES USED

Azure Data Lake Storage Gen2 (ADLS Gen2)

ADLS Gen2 serves as the primary cloud storage layer for the pipeline, providing scalable and secure hierarchical storage for structured and unstructured big data. It stores the dataset across raw, silver, and gold containers, enabling organised separation between unprocessed, cleaned, and aggregated data layers.

Azure Data Factory (ADF)

Azure Data Factory acts as the orchestration and workflow automation service within the pipeline. It coordinates data movement, triggers Databricks notebooks sequentially, and monitors execution flow to ensure the pipeline runs in the correct order without manual intervention.

Apache Spark & Azure Databricks

Apache Spark enables distributed in-memory processing for large-scale ETL operations, making it ideal for transforming dirty datasets efficiently. Azure Databricks provides a managed Spark environment where PySpark notebooks perform schema standardisation, cleaning, deduplication, missing value handling, and feature engineering.

Azure Synapse Analytics

Azure Synapse Analytics provides serverless SQL-based analytical querying over the transformed Parquet files stored in ADLS. Using OPENROWSET, Synapse queries the Silver and Gold datasets directly without requiring database loading, enabling fast and cost-efficient business analysis.

Microsoft Power BI

Power BI serves as the business intelligence and visualisation layer of the pipeline. It converts processed analytical data into interactive dashboards, charts, and KPI cards, allowing users to explore learner performance, engagement, and behavioural insights visually.

03 - DATASET

Online Learning Analytics Dataset

Injected Data Quality Issues

The following issues were deliberately introduced to replicate what arrives from real survey tools, learning management systems, and mobile apps:

Missing Values ~219 nulls	Median imputation for hours, quiz, satisfaction, login; 'Unknown' for country
Duplicate Rows 17 rows	Removed via dropDuplicates() — keeps one canonical record per user

Inconsistent Labels ~290 rows	Gender/Education/Device/Completion standardised to canonical forms using PySpark when()
Outliers ~27 rows	Domain-bound filters: age 10–100, hours 0–168, quiz 0–100, satisfaction 1–5
Whitespace / Format ~40 rows	F.trim() applied to all string columns; case normalised with F.lower()
Wrong Data Types All cols	Explicit casting to IntegerType / DoubleType after cleaning
Label Case Mixing ~50 rows	Lowercase comparison before matching removes MALE/Male/male ambiguity

online_learning_dirty

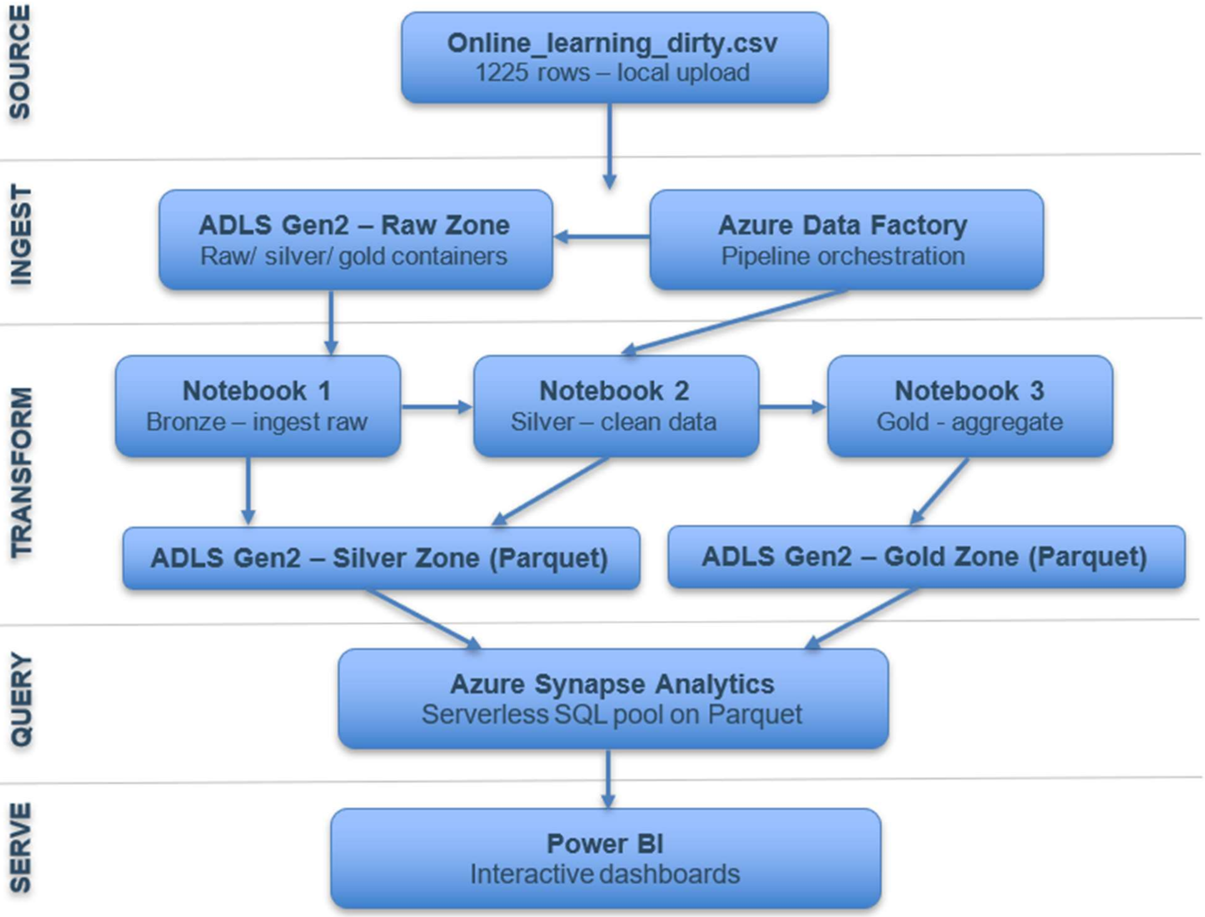
user_id	age	gender	country	education_level	hours_spent_weekly	preferred_device	course_completed	satisfaction_rating	login_frequency	quiz_score_avg
user_0598	33	Non-binary	USA	PhD	10.3	Desktop	No	4	2	74.3
user_0716	55	male	Canada	PhD	19.7	Tablet	TRUE	4	9	92.1
user_0922	29	Other	Canada	Master	9.2	Desktop	No	1	5	78.7
user_1160	36	Other	USA	Master	16.9	Desktop	Yes	1	4	69.4
user_0416	46	Male	Canada	Other	6.9	Tablet	No	3	11	86.0
user_0169	54	Other	Canada	Bachelor	18.9	Desktop	No	4	1	87.4
user_1189	27	Female	Canada	master	7.8	Tablet	FALSE	3	8	46.7
user_1104	59	Male	Australia	PhD	200	Desktop	Yes	1	3	73.1
user_0412	43	Other	Canada	Master	1.3	Mobile	Yes	3	6	47.7
user_0102	25	Male	UK	Master	5.4	Desktop	Yes	4	14	72.4
user_0241	43	Female	UK	Other	11.4	Tablet	No		9	49.8
user_0156	42	Female	India	Bachelor	12.2	Desktop	Yes	1	4	55.2
user_0356	33	Male	UK	Other	11.1	Tablet	Yes	5	11	51.5
user_0334	19	Female	India	PhD	-10	Desktop	Yes	3	13	59.6
user_0364	49	Other	USA	High School		Desktop	Yes	3	5	95.9
user_0050	31	Male		PhD	12.3	Mobile	No	3	8	40.4
user_0782	43	Male	Australia	High School		Mobile	No	2	14	

04 – METHODOLOGY

Pipeline Architecture & Azure Stack

The pipeline follows the Medallion Architecture — an industry-standard pattern that organizes data into three progressive quality layers. Each layer builds on the previous one, ensuring data quality, maintainability, and clear separation of concerns. The architecture maps directly to Azure's native services.

5-Step Pipeline Flow



Medallion Architecture Layers

Raw Zone	Silver Zone	Gold Zone
Original dirty CSV stored as-is in ADLS Gen2. Nothing is modified - this is your audit baseline. Data arrives here directly via ADF copy activity.	Cleaned, deduplicated Parquet files partitioned by education_level. All 7 issue types resolved here by the Databricks PySpark notebook.	Aggregated business metrics partitioned by dimension (device, country, education, satisfaction). Directly consumed by Power BI dashboards.

05 – IMPLEMENTATION

STAGE 1 — SOURCE

This stage represents the origin of the pipeline where the raw dataset is introduced before any cloud processing begins. The dataset contains intentionally injected errors to simulate real-world dirty data conditions.

- Source file: online_learning_dirty.csv
- Contains 1,225 raw records
- Includes missing values, duplicates, inconsistent labels, and outliers
- Serves as the unprocessed input before Azure ingestion

STAGE 2 — INGEST

In this stage, the raw dataset is uploaded into Azure storage and the workflow orchestration process begins. Data Factory coordinates the movement and triggering of downstream pipeline activities.

- ADLS Gen2 stores all pipeline data
- Uses three containers: raw/, silver/, gold/
- Raw CSV uploaded unchanged into raw/ container
- Azure Data Factory automates workflow execution
- Triggers Databricks notebooks sequentially

STAGE 3 — TRANSFORM

This stage performs all cleaning, preprocessing, and feature engineering operations using Azure Databricks with PySpark. Raw data is converted into structured and analysis-ready format.

- Managed through Azure Databricks notebooks
- Bronze notebook standardises schema and converts CSV to Parquet
- Silver notebook performs data cleaning and validation
- Removes duplicates, nulls, outliers, and formatting issues
- Creates engineered fields:
- engagement_score
- satisfaction_group
- completed_binary

STAGE 4 — QUERY

This stage enables analytical querying of processed data using Azure Synapse Analytics. SQL is used to derive insights directly from Parquet files stored in ADLS.

- Uses Azure Synapse serverless SQL pool
- Reads Parquet files via OPENROWSET
- Queries Silver and Gold layer data directly
- Generates analytics on performance, completion, and satisfaction
- Improves speed using partitioned/aggregated datasets

STAGE 5 — SERVE

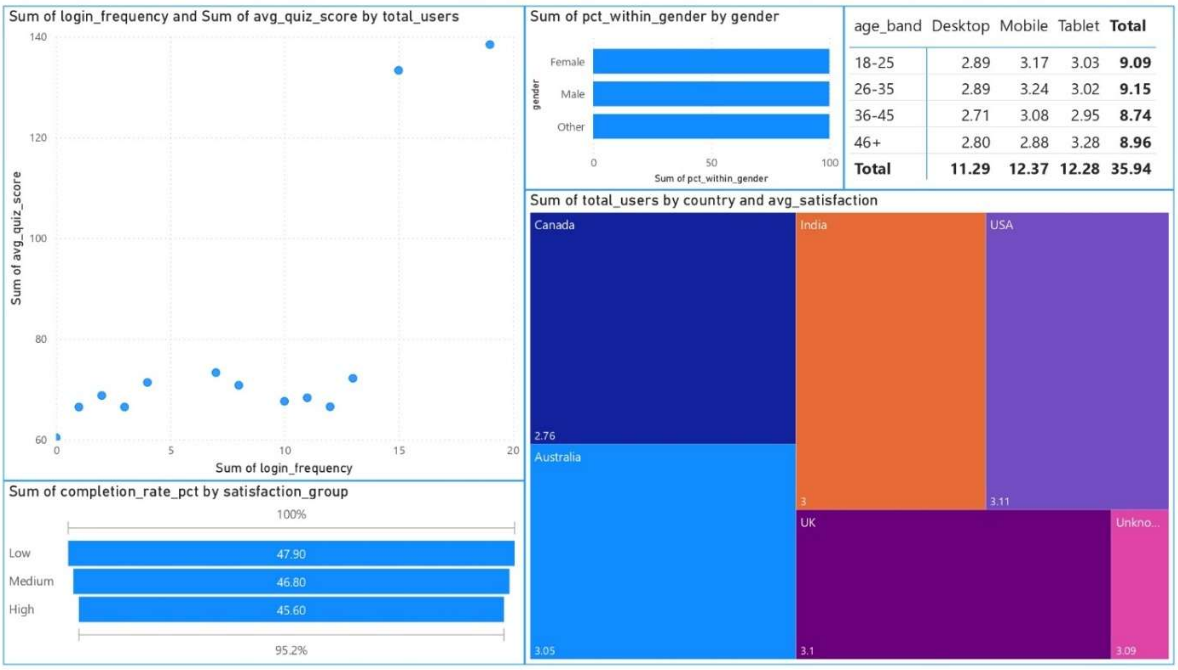
This is the final stage where processed analytical data is converted into business intelligence dashboards. Power BI visualises insights for end-user interpretation and decision-making.

- Power BI used for dashboard creation
- Imports Synapse SQL query outputs
- Displays KPI cards and analytical charts
- Provides interactive learner performance insights
- Acts as final presentation layer of the pipeline

06 – MICROSOFT POWER BI OUTPUT

Executive Performance Dashboard: Provides high-level summary metrics and overall learner performance trends.

Behavioural & Advanced Analytics Dashboard: Provides deeper analytical insights into engagement, demographic patterns, and behavioural relationships.



07 - FUTURE WORK

Roadmap & Planned Enhancements

The current pipeline provides a solid batch-processing foundation. The following enhancements represent a natural progression from student project to production-grade system, organized by implementation complexity and timeline:

Phase	Enhancement	Azure Tool	Description
Short-term	**Real-Time Streaming**	*Azure Event Hub + Stream Analytics*	Ingest live clickstream and session data to replace batch CSV uploads
Short-term	**Automated Orchestration**	*Azure Data Factory Triggers*	Replace manual pipeline runs with scheduled or event-driven triggers
Mid-term	**Machine Learning**	*Azure Machine Learning*	Train a completion-prediction model on gold-zone features using AutoML
Mid-term	**Predictive Dashboards**	*Power BI + Azure ML*	Embed predicted course completion probabilities into Power BI visuals
Long-term	**Anomaly Detection**	*Azure Anomaly Detector API*	Automatically flag unusual engagement patterns without manual rules
Long-term	**Multi-Source Scaling**	*ADLS Gen2 + ADF Connectors*	Integrate external APIs, IoT devices, and secondary learning platforms

Conclusion

This project demonstrates the full data engineering lifecycle — from dirty, inconsistent raw data to clean, queryable, visualized business insights — entirely on Microsoft Azure. The Medallion Architecture (Raw → Silver → Gold) provides a robust, auditable, and scalable foundation that mirrors how production data platforms are built at enterprise scale.

By deliberately introducing real-world data quality problems and systematically resolving them through PySpark transformations, the pipeline validates that Azure Databricks + ADLS Gen2 + Synapse Analytics + Power BI is a cost-effective, production-capable stack accessible even under a student credit budget.